TASK 10 – DOCUMENTATION & SUBMISSION

Data Set:

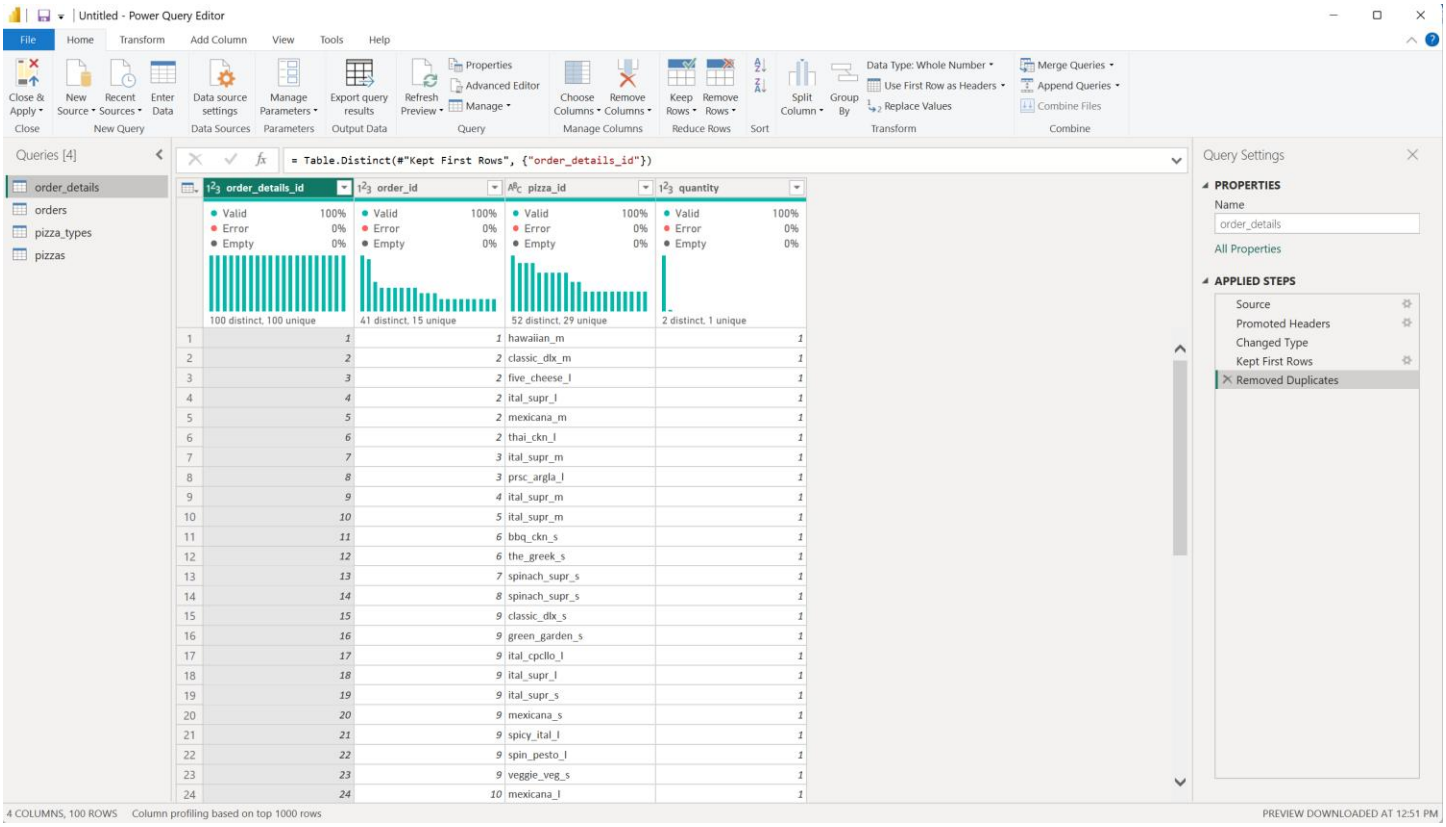
Pizza Sales

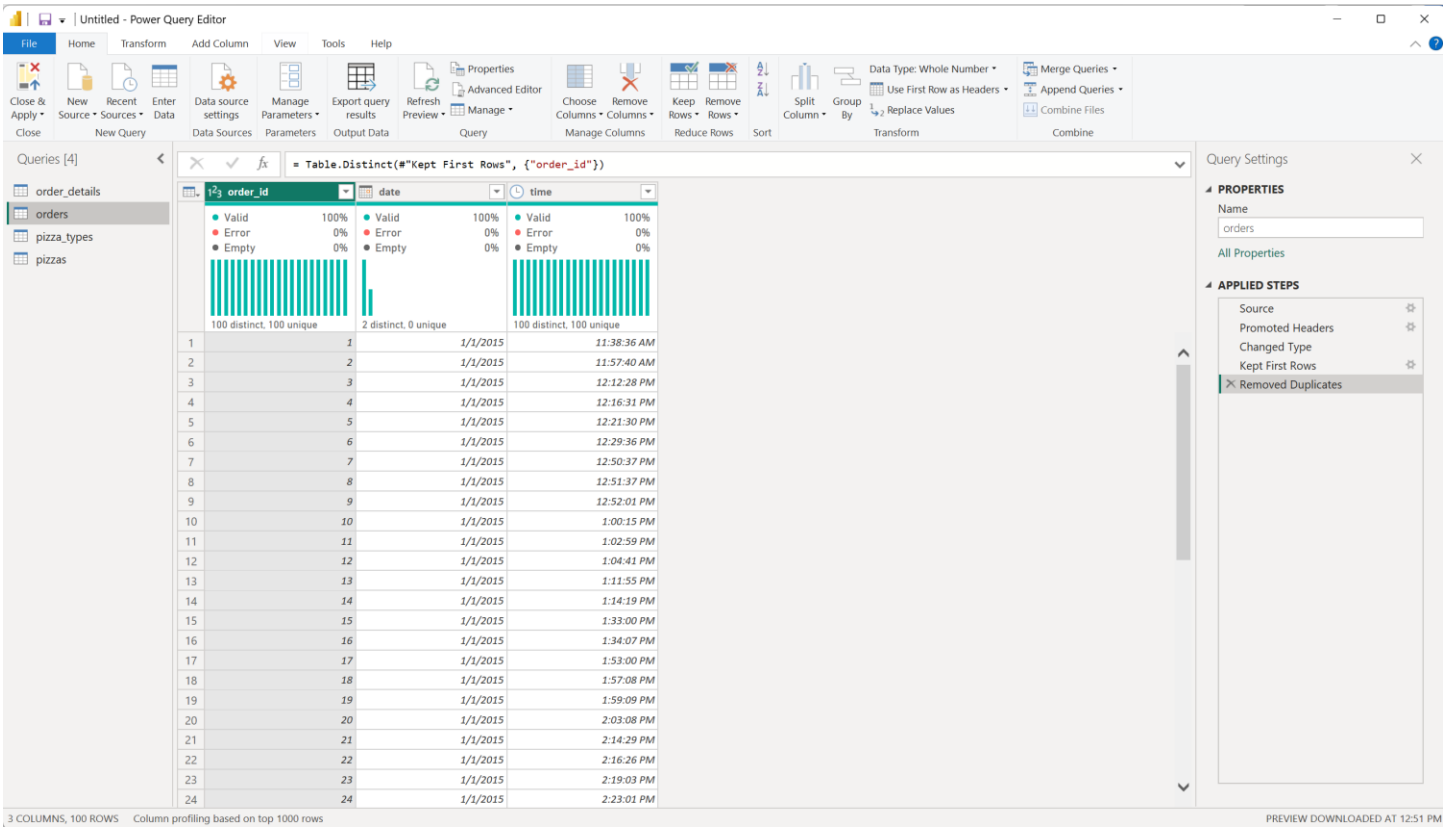
The pizza sales dataset, capturing customer orders, order details, pizza attributes, and category classifications.

1. Orders Table
    a. Captures each customer transaction with a unique order ID.
    b. Records the date and time when the order was placed.
    c. Serves as the primary entry point for tracking sales activity.
2. Order Details Table
    a. Provides a line-level breakdown of pizzas within each order.
    b. Links to the orders table through the foreign key order_id.
    c. Associates each pizza with its pizza_id and specifies the quantity ordered.
    d. Enables detailed analysis of order composition and item frequency.
3. Pizzas Table
    a. Defines each pizza variant by a unique pizza_id.
    b. Includes size options (Small, Medium, Large, X Large, XX Large).
    c. Stores the price in USD, allowing revenue calculations.
    d. Connects to pizza_types for broader classification.
4. Pizza Types Table
    a. Contains descriptive attributes of each pizza type.
    b. Provides the name as shown on the menu.
    c. Categorizes pizzas into Classic, Chicken, Supreme, or Veggie.
    d. Lists ingredients in a comma-delimited format, supporting menu analysis.

Data Cleaning:

DETECT AND REMOVE DUPLICATES OR MISSING VALUES





Untitled - Power Query Editor

File | Home | Transform | Add Column | View | Tools | Help

Close & Apply | New Source | Recent Sources | Enter Data | Data source settings | Manage Parameters | Export query results | Refresh Preview | Properties | Advanced Editor | Manage | Choose Columns | Remove Columns | Keep Rows | Remove Rows | Split Column | Group By | Data Type: Text | Use First Row as Headers | Replace Values | Merge Queries | Append Queries | Combine Files

Queries [4]

- order_details
- orders
- pizza_types
- pizzas

```
= Table.Distinct(#"Changed Type1", {"pizza_type_id"})
```

	pizza_type_id	name	category	ingredients
	Valid 100%	Valid 100%	Valid 100%	Valid 100%
	Error 0%	Error 0%	Error 0%	Error 0%
	Empty 0%	Empty 0%	Empty 0%	Empty 0%
	32 distinct, 32 unique	32 distinct, 32 unique	4 distinct, 0 unique	32 distinct, 32 unique
1	bbq_ckn	The Barbecue Chicken Pizza	Chicken	Barbecued Chicken, Red Peppers, Green Peppers, Tomatoes, Red Onio...
2	cali_ckn	The California Chicken Pizza	Chicken	Chicken, Artichoke, Spinach, Garlic, Jalapeno Peppers, Fontina Cheese,...
3	ckn_alfredo	The Chicken Alfredo Pizza	Chicken	Chicken, Red Onions, Red Peppers, Mushrooms, Asiago Cheese, Alfred...
4	ckn_pesto	The Chicken Pesto Pizza	Chicken	Chicken, Tomatoes, Red Peppers, Spinach, Garlic, Pesto Sauce
5	southw_ckn	The Southwest Chicken Pizza	Chicken	Chicken, Tomatoes, Red Peppers, Red Onions, Jalapeno Peppers, Corn,...
6	thai_ckn	The Thai Chicken Pizza	Chicken	Chicken, Pineapple, Tomatoes, Red Peppers, Thai Sweet Chilli Sauce
7	big_meat	The Big Meat Pizza	Classic	Bacon, Pepperoni, Italian Sausage, Chorizo Sausage
8	classic_dlx	The Classic Deluxe Pizza	Classic	Pepperoni, Mushrooms, Red Onions, Red Peppers, Bacon
9	hawaiian	The Hawaiian Pizza	Classic	Sliced Ham, Pineapple, Mozzarella Cheese
10	ital_cpclio	The Italian Capocollo Pizza	Classic	Capocollo, Red Peppers, Tomatoes, Goat Cheese, Garlic, Oregano
11	napolitana	The Napolitana Pizza	Classic	Tomatoes, Anchovies, Green Olives, Red Onions, Garlic
12	pep_msh_pep	The Pepperoni, Mushroom, and Peppers Pizza	Classic	Pepperoni, Mushrooms, Green Peppers
13	pepperoni	The Pepperoni Pizza	Classic	Mozzarella Cheese, Pepperoni
14	the_greek	The Greek Pizza	Classic	Kalamata Olives, Feta Cheese, Tomatoes, Garlic, Beef Chuck Roast, Re...
15	brie_carre	The Brie Carre Pizza	Supreme	Brie Carre Cheese, Prosciutto, Caramelized Onions, Pears, Thyme, Garlic
16	calabrese	The Calabrese Pizza	Supreme	'Nduja Salami, Pancetta, Tomatoes, Red Onions, Friggitello Peppers, G...
17	ital_supr	The Italian Supreme Pizza	Supreme	Calabrese Salami, Capocollo, Tomatoes, Red Onions, Green Olives, Gar...
18	peppr_salami	The Pepper Salami Pizza	Supreme	Genoa Salami, Capocollo, Pepperoni, Tomatoes, Asiago Cheese, Garlic
19	prsc_argla	The Prosciutto and Arugula Pizza	Supreme	Prosciutto di San Daniele, Arugula, Mozzarella Cheese
20	sicilian	The Sicilian Pizza	Supreme	Coarse Sicilian Salami, Tomatoes, Green Olives, Luganega Sausage, Oni...
21	soppressata	The Soppressata Pizza	Supreme	Soppressata Salami, Fontina Cheese, Mozzarella Cheese, Mushrooms, ...
22	spicy_ital	The Spicy Italian Pizza	Supreme	Capocollo, Tomatoes, Goat Cheese, Artichokes, Peperoncini verdi, Garlic
23	spinach_supr	The Spinach Supreme Pizza	Supreme	Spinach, Red Onions, Pepperoni, Tomatoes, Artichokes, Kalamata Oliv...
24	five_cheese	The Five Cheese Pizza	Veggie	Mozzarella Cheese, Provolone Cheese, Smoked Gouda Cheese, Roman...

Query Settings

PROPERTIES

Name: pizza_types

All Properties

APPLIED STEPS

- Source
- Changed Type
- Promoted Headers
- Changed Type1
- Removed Duplicates

4 COLUMNS, 32 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 12:51 PM

Untitled - Power Query Editor

File | Home | Transform | Add Column | View | Tools | Help

Close & Apply | New Source | Recent Sources | Enter Data | Data source settings | Manage Parameters | Export query results | Refresh Preview | Properties | Advanced Editor | Manage | Choose Columns | Remove Columns | Keep Rows | Remove Rows | Split Column | Group By | Data Type: Text | Use First Row as Headers | Replace Values | Merge Queries | Append Queries | Combine Files

Queries [4]

- order_details
- orders
- pizza_types
- pizzas

```
= Table.Distinct(#"Changed Type", {"pizza_id"})
```

	pizza_id	pizza_type_id	size	price
	Valid 100%	Valid 100%	Valid 100%	Valid 100%
	Error 0%	Error 0%	Error 0%	Error 0%
	Empty 0%	Empty 0%	Empty 0%	Empty 0%
	96 distinct, 96 unique	32 distinct, 1 unique	5 distinct, 2 unique	27 distinct, 16 unique
1	bbq_ckn_s	bbq_ckn	S	12.75
2	bbq_ckn_m	bbq_ckn	M	16.75
3	bbq_ckn_l	bbq_ckn	L	20.75
4	cali_ckn_s	cali_ckn	S	12.75
5	cali_ckn_m	cali_ckn	M	16.75
6	cali_ckn_l	cali_ckn	L	20.75
7	ckn_alfredo_s	ckn_alfredo	S	12.75
8	ckn_alfredo_m	ckn_alfredo	M	16.75
9	ckn_alfredo_l	ckn_alfredo	L	20.75
10	ckn_pesto_s	ckn_pesto	S	12.75
11	ckn_pesto_m	ckn_pesto	M	16.75
12	ckn_pesto_l	ckn_pesto	L	20.75
13	southw_ckn_s	southw_ckn	S	12.75
14	southw_ckn_m	southw_ckn	M	16.75
15	southw_ckn_l	southw_ckn	L	20.75
16	thai_ckn_s	thai_ckn	S	12.75
17	thai_ckn_m	thai_ckn	M	16.75
18	thai_ckn_l	thai_ckn	L	20.75
19	big_meat_s	big_meat	S	12
20	big_meat_m	big_meat	M	16
21	big_meat_l	big_meat	L	20.5
22	classic_dlx_s	classic_dlx	S	12
23	classic_dlx_m	classic_dlx	M	16
24	classic_dlx_l	classic_dlx	L	20.5

Query Settings

PROPERTIES

Name: pizzas

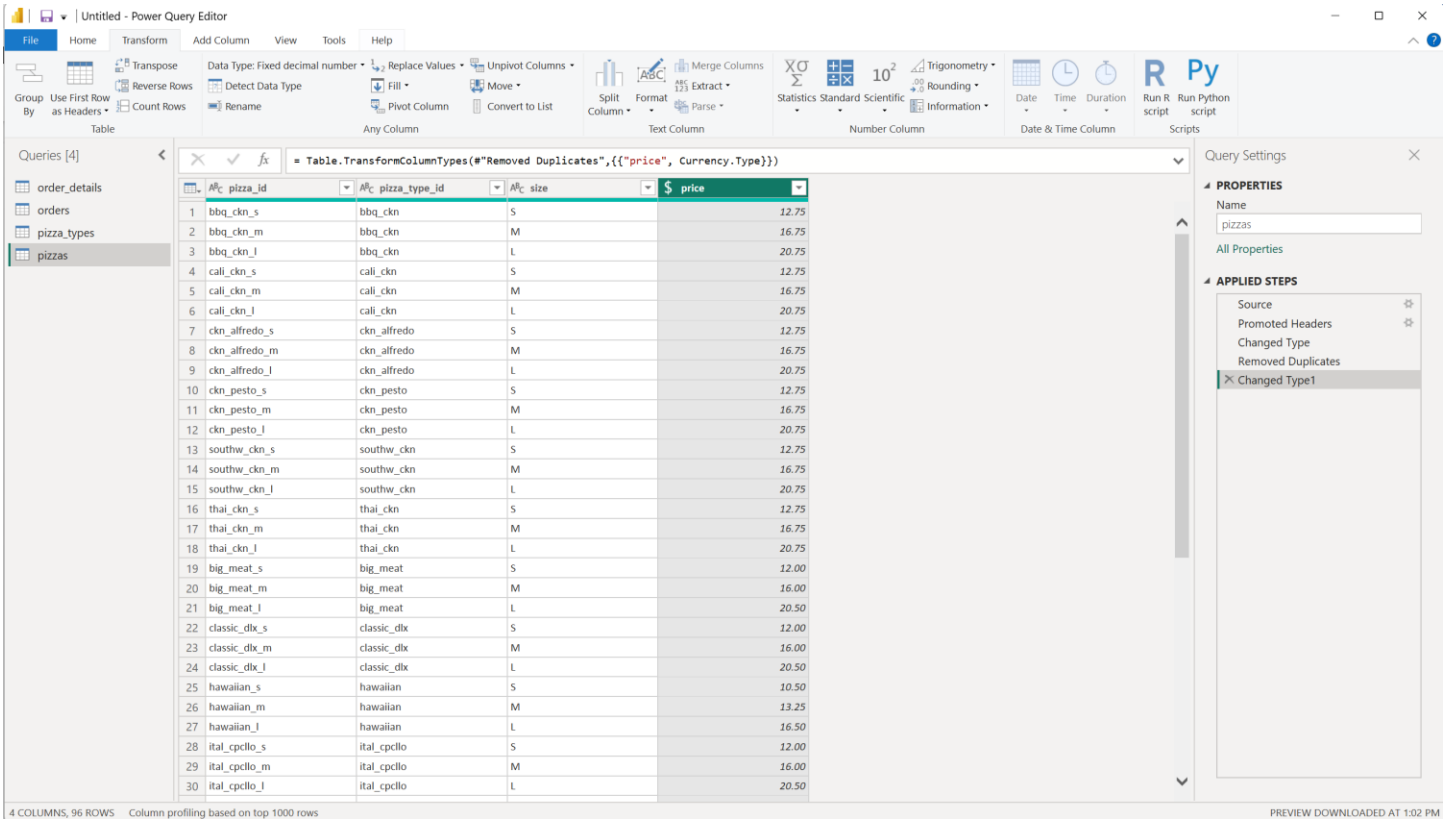
All Properties

APPLIED STEPS

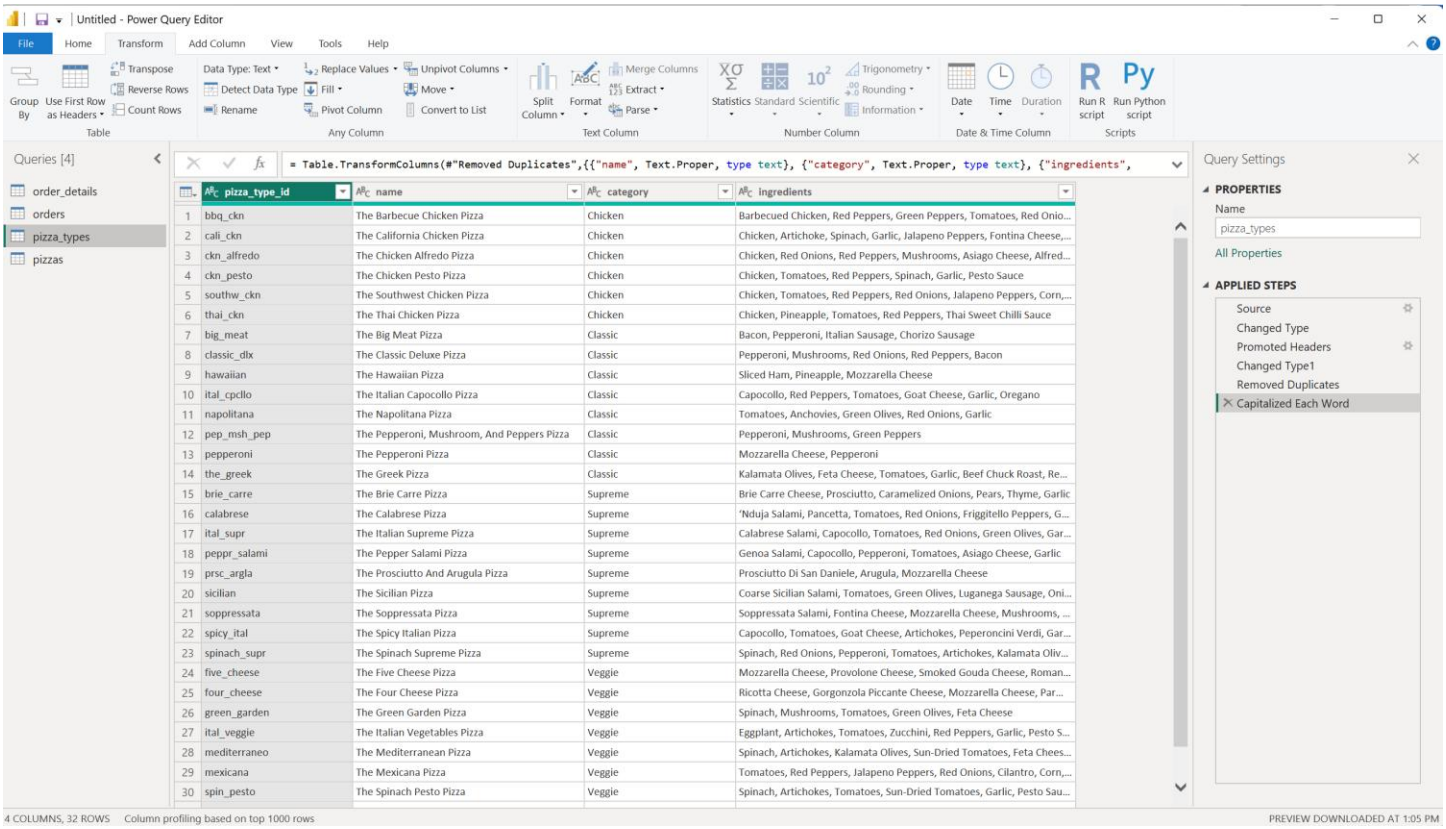
- Source
- Promoted Headers
- Changed Type
- Removed Duplicates

4 COLUMNS, 96 ROWS Column profiling based on top 1000 rows PREVIEW DOWNLOADED AT 12:51 PM

STANDARDIZE COLUMN NAMES AND DATA TYPES



	pizza_id	pizza_type_id	size	price
1	bbq_ckn_s	bbq_ckn	S	12.75
2	bbq_ckn_m	bbq_ckn	M	16.75
3	bbq_ckn_l	bbq_ckn	L	20.75
4	cali_ckn_s	cali_ckn	S	12.75
5	cali_ckn_m	cali_ckn	M	16.75
6	cali_ckn_l	cali_ckn	L	20.75
7	ckn_alfredo_s	ckn_alfredo	S	12.75
8	ckn_alfredo_m	ckn_alfredo	M	16.75
9	ckn_alfredo_l	ckn_alfredo	L	20.75
10	ckn_pesto_s	ckn_pesto	S	12.75
11	ckn_pesto_m	ckn_pesto	M	16.75
12	ckn_pesto_l	ckn_pesto	L	20.75
13	southw_ckn_s	southw_ckn	S	12.75
14	southw_ckn_m	southw_ckn	M	16.75
15	southw_ckn_l	southw_ckn	L	20.75
16	thai_ckn_s	thai_ckn	S	12.75
17	thai_ckn_m	thai_ckn	M	16.75
18	thai_ckn_l	thai_ckn	L	20.75
19	big_meat_s	big_meat	S	12.00
20	big_meat_m	big_meat	M	16.00
21	big_meat_l	big_meat	L	20.50
22	classic_dlx_s	classic_dlx	S	12.00
23	classic_dlx_m	classic_dlx	M	16.00
24	classic_dlx_l	classic_dlx	L	20.50
25	hawaiian_s	hawaiian	S	10.50
26	hawaiian_m	hawaiian	M	13.25
27	hawaiian_l	hawaiian	L	16.50
28	ital_cpcllo_s	ital_cpcllo	S	12.00
29	ital_cpcllo_m	ital_cpcllo	M	16.00
30	ital_cpcllo_l	ital_cpcllo	L	20.50



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7	big_meat	The Big Meat Pizza	Classic	Bacon, Pepperoni, Italian Sausage, Chorizo Sausage
8	classic_dlx	The Classic Deluxe Pizza	Classic	Pepperoni, Mushrooms, Red Onions, Red Peppers, Bacon
9	hawaiian	The Hawaiian Pizza	Classic	Sliced Ham, Pineapple, Mozzarella Cheese
10	ital_cpcllo	The Italian Capocollo Pizza	Classic	Capocollo, Red Peppers, Tomatoes, Goat Cheese, Garlic, Oregano
11	napolitana	The Napolitana Pizza	Classic	Tomatoes, Anchovies, Green Olives, Red Onions, Garlic
12	pep_msh_pep	The Pepperoni, Mushroom, And Peppers Pizza	Classic	Pepperoni, Mushrooms, Green Peppers
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14	the_greek	The Greek Pizza	Classic	Kalamata Olives, Feta Cheese, Tomatoes, Garlic, Beef Chuck Roast, Re...
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18	peppr_salami	The Pepper Salami Pizza	Supreme	Genoa Salami, Capocollo, Pepperoni, Tomatoes, Asiago Cheese, Garlic
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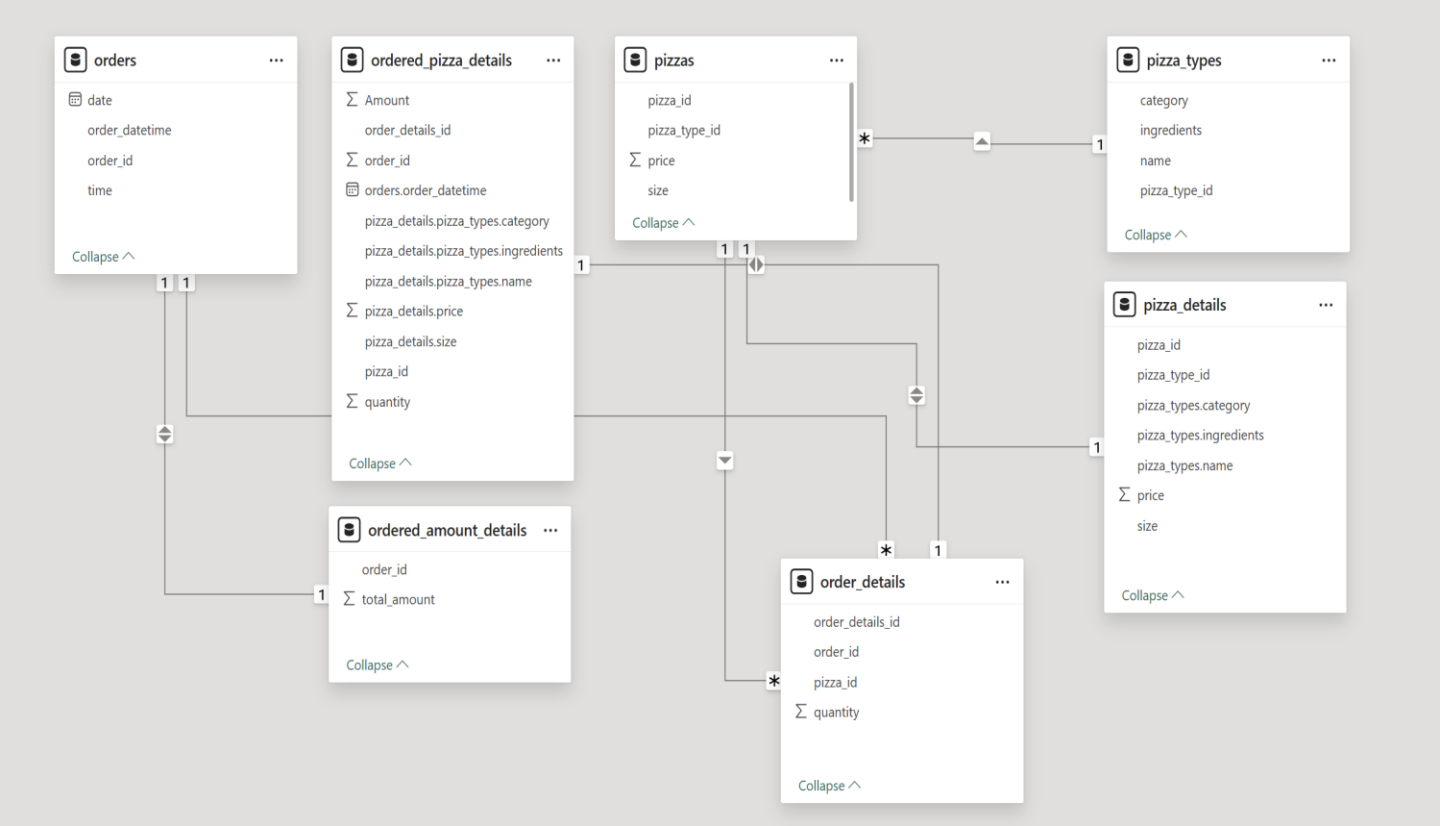
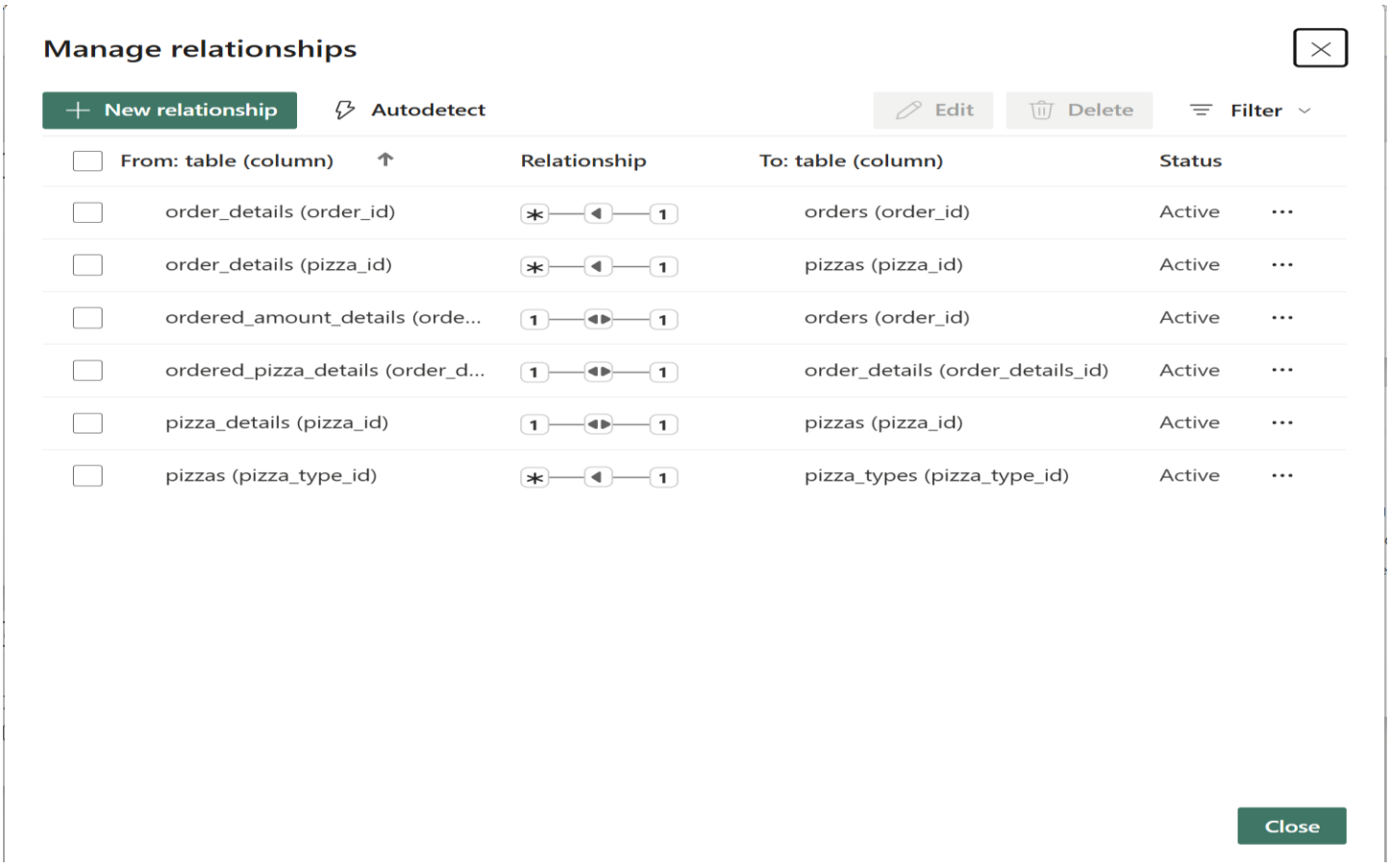
24	five_cheese	The Five Cheese Pizza	Veggie	Mozzarella Cheese, Provolone Cheese, Smoked Gouda Cheese, Roman...
25	four_cheese	The Four Cheese Pizza	Veggie	Ricotta Cheese, Gorgonzola Piccante Cheese, Mozzarella Cheese, Par...
26	green_garden	The Green Garden Pizza	Veggie	Spinach, Mushrooms, Tomatoes, Green Olives, Feta Cheese
27	ital_veggie	The Italian Vegetables Pizza	Veggie	Eggplant, Artichokes, Tomatoes, Zucchini, Red Peppers, Garlic, Pesto S...
28	mediterraneo	The Mediterranean Pizza	Veggie	Spinach, Artichokes, Kalamata Olives, Sun-Dried Tomatoes, Feta Chees...
29	mexicana	The Mexicana Pizza	Veggie	Tomatoes, Red Peppers, Jalapeno Peppers, Red Onions, Cilantro, Corn,...
30	spin_pesto	The Spinach Pesto Pizza	Veggie	Spinach, Artichokes, Tomatoes, Sun-Dried Tomatoes, Garlic, Pesto Sau...

DATA MODELING:





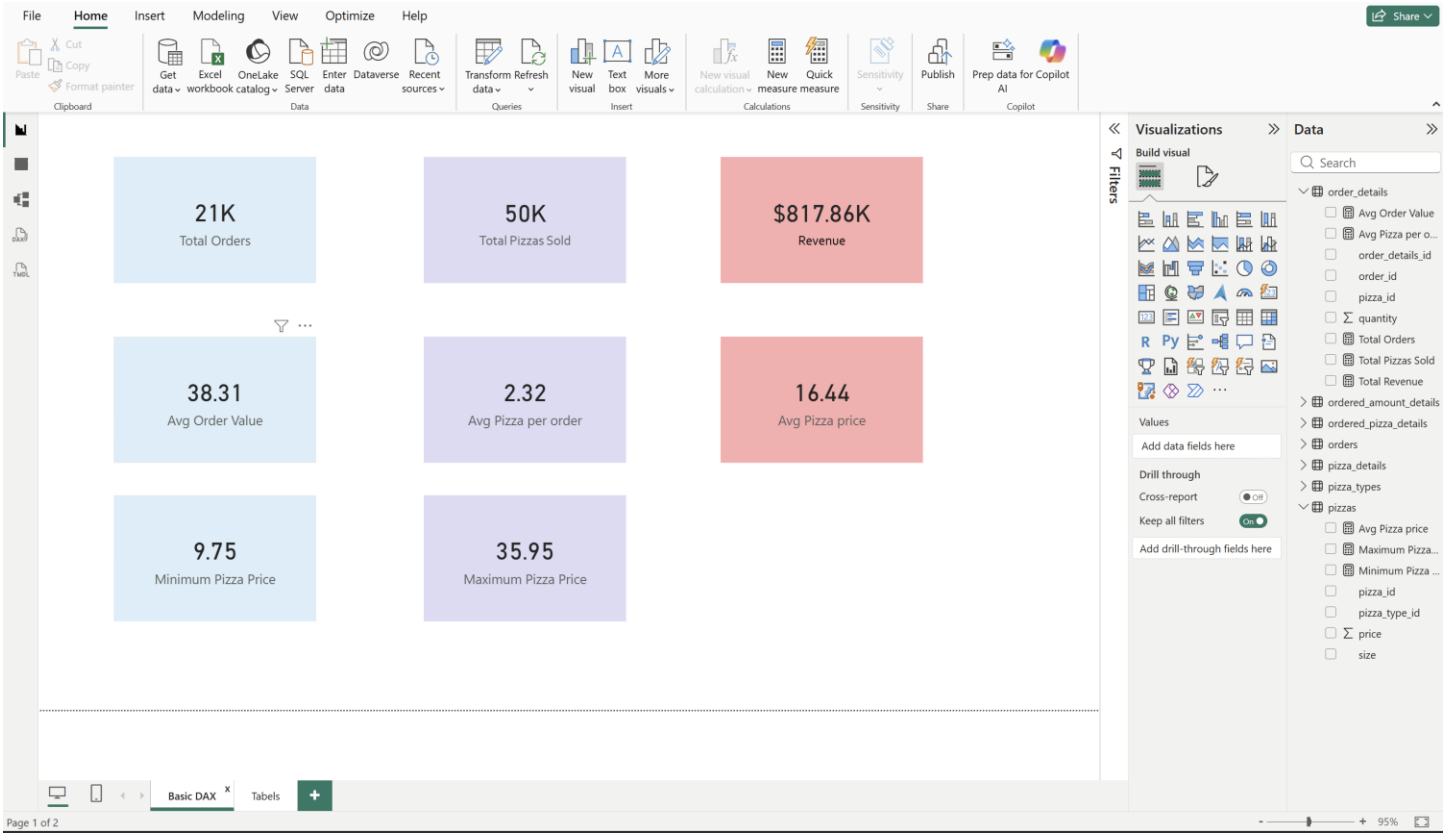
Manage relationships

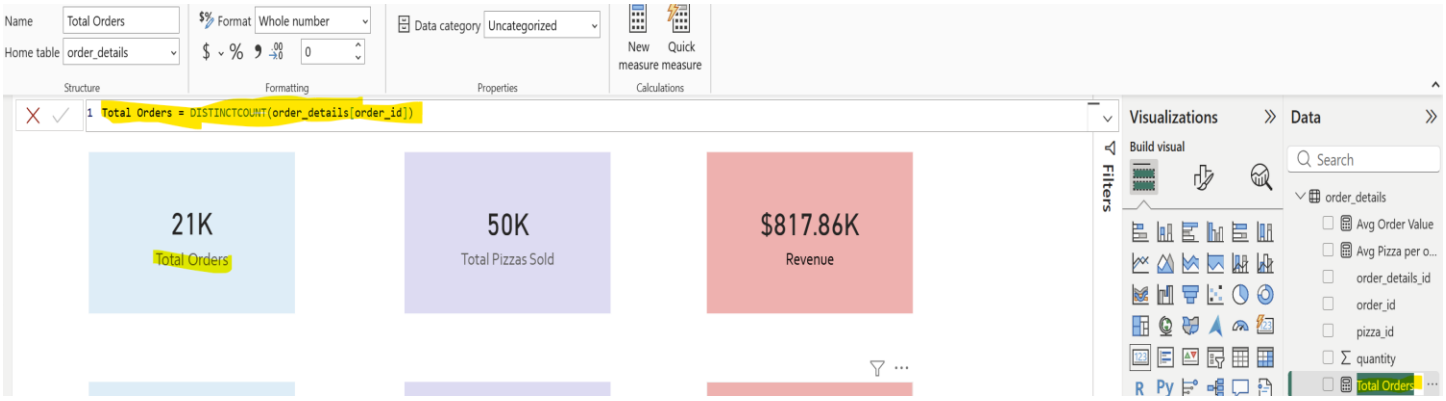
New relationship | Autodetect | Edit | Delete | Filter

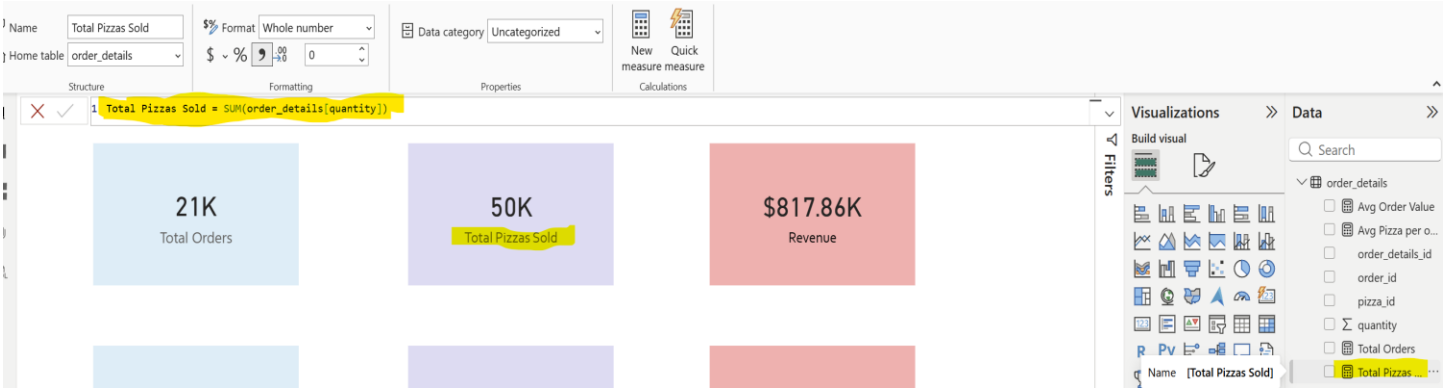
From: table (column)	Relationship	To: table (column)	Status
order_details (order_id)	* — ◀ — 1	orders (order_id)	Active
order_details (pizza_id)	* — ◀ — 1	pizzas (pizza_id)	Active
ordered_amount_details (orde...	1 — ◀▶ — 1	orders (order_id)	Active
ordered_pizza_details (order_d...	1 — ◀▶ — 1	order_details (order_details_id)	Active
pizza_details (pizza_id)	1 — ◀▶ — 1	pizzas (pizza_id)	Active
pizzas (pizza_type_id)	* — ◀ — 1	pizza_types (pizza_type_id)	Active

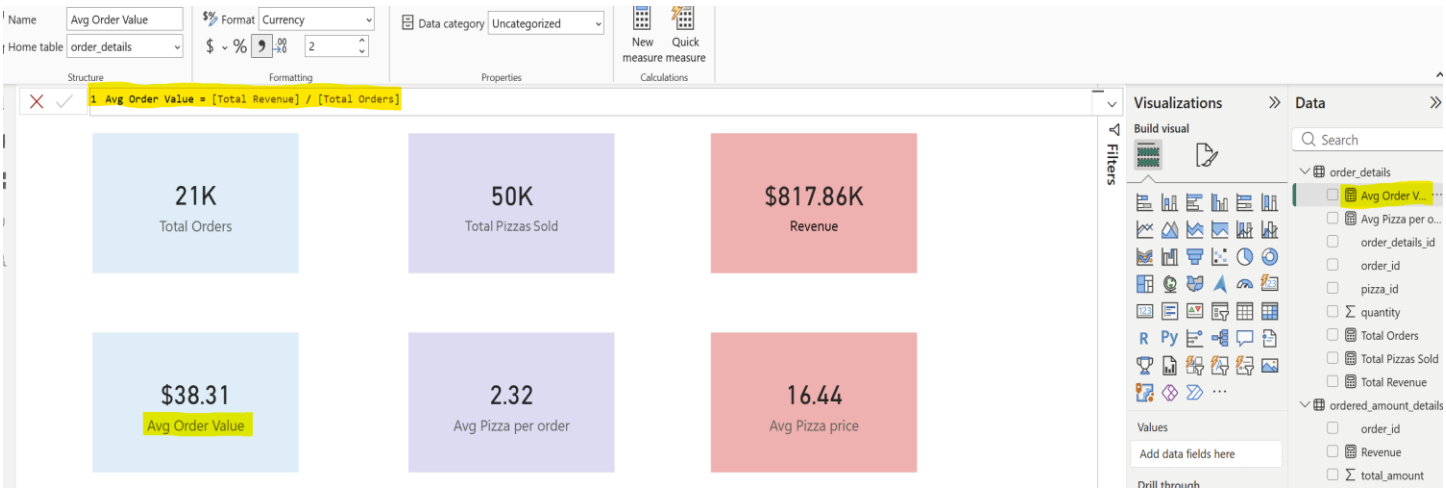
Close

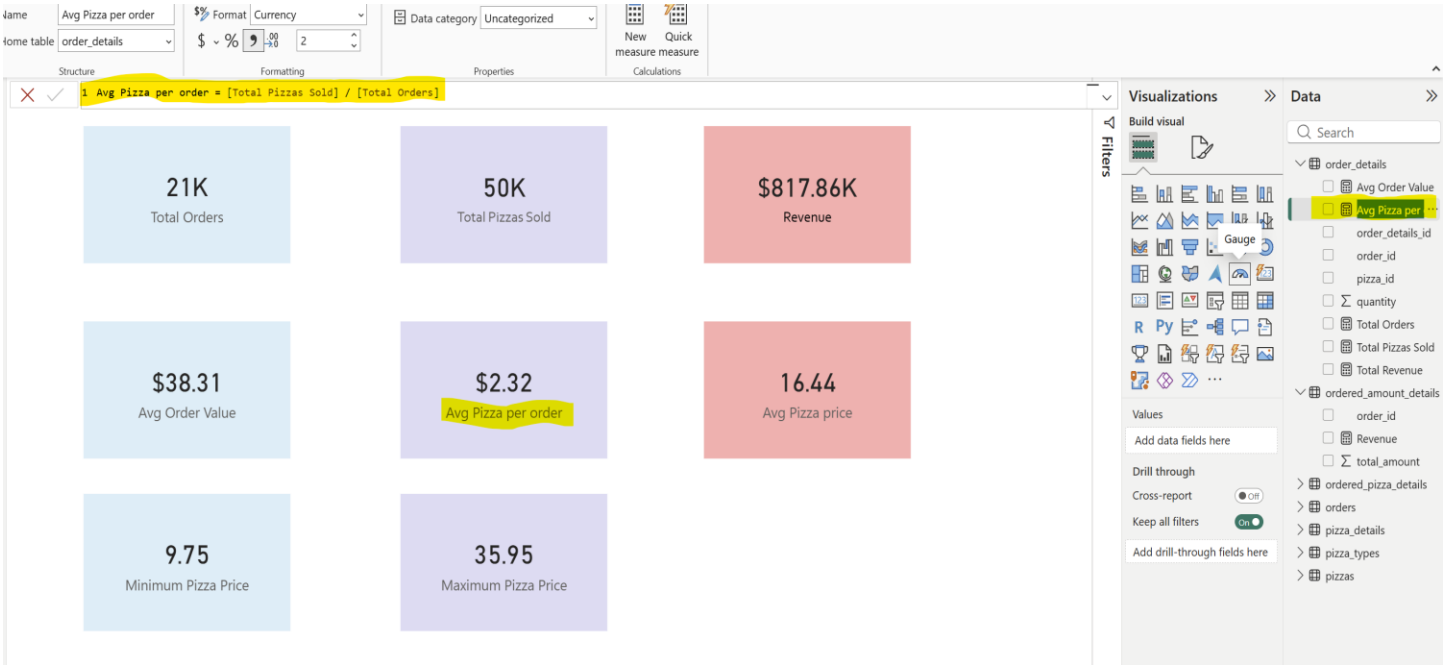
DAX MEASURES:

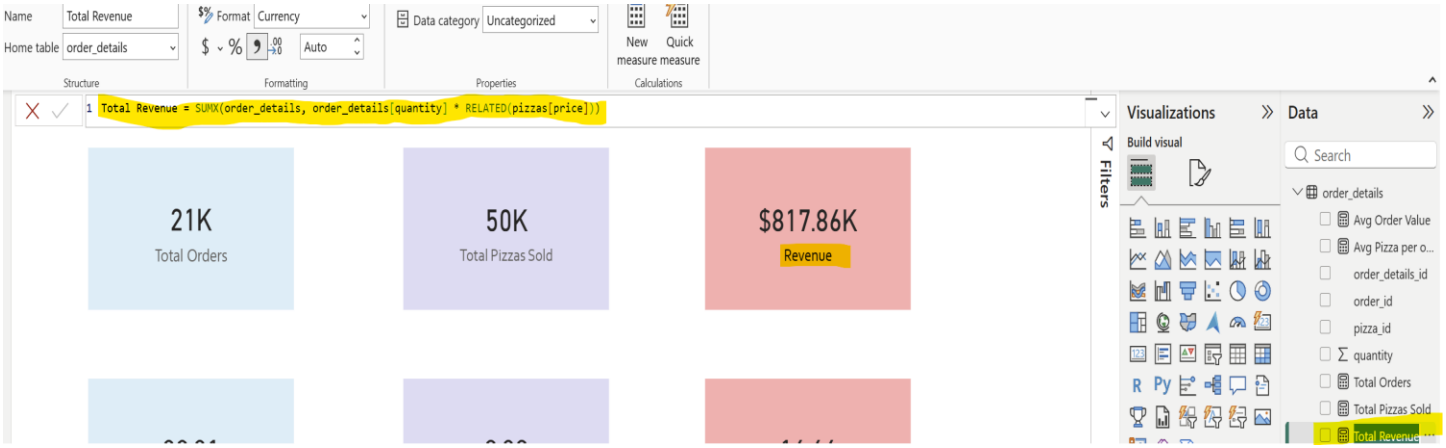


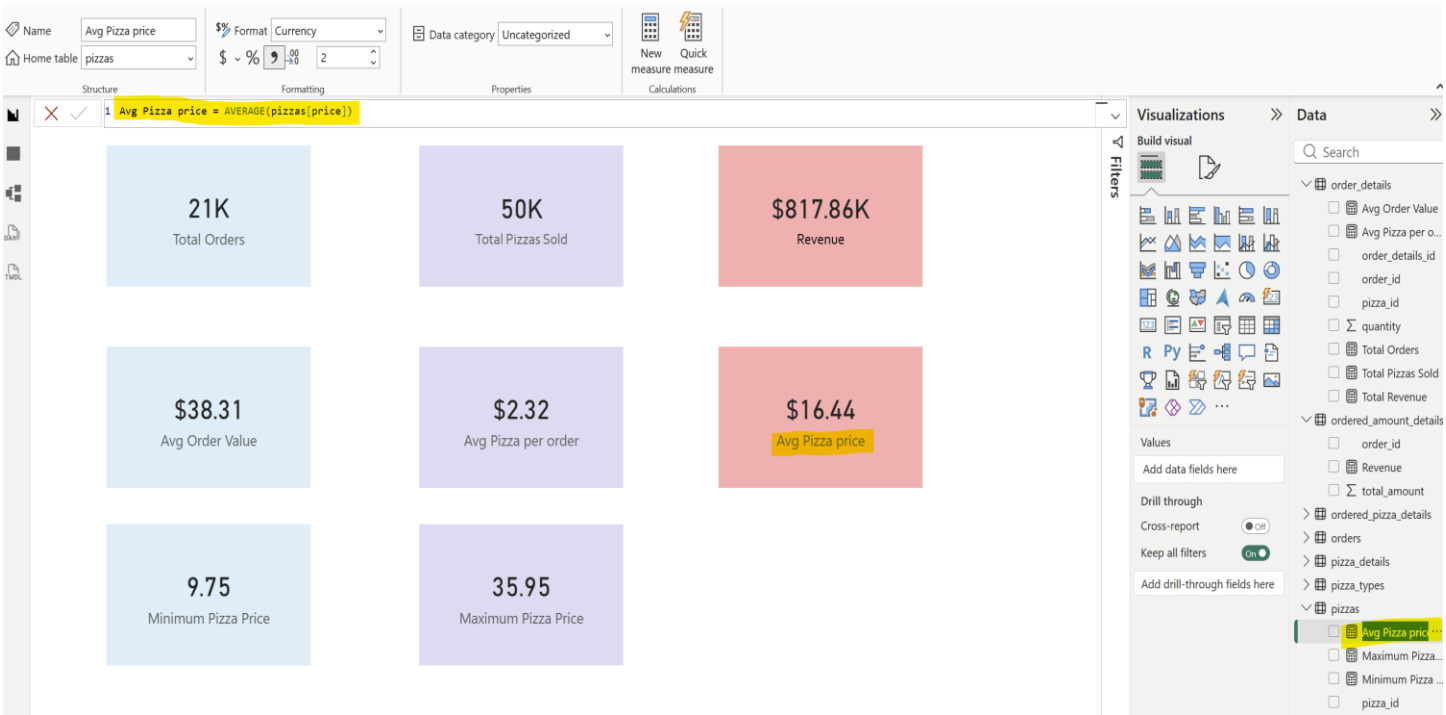





Name Avg Order Value
Home table order_details
Structure
Format Currency
Formatting
Data category Uncategorized
Properties
New measure
Quick measure
Calculations
1 Avg Order Value = [Total Revenue] / [Total Orders]
21K
Total Orders
50K
Total Pizzas Sold
$817.86K
Revenue
$38.31
Avg Order Value
2.32
Avg Pizza per order
16.44
Avg Pizza price
Visualizations
Build visual
Values
Add data fields here
Drill through
Filters
Data
Search
order_details
Avg Order V...
Avg Pizza per o...
order_details_id
order_id
pizza_id
quantity
Total Orders
Total Pizzas Sold
Total Revenue
ordered_amount_details
order_id
Revenue
total_amount


Name Avg Pizza per order
Home table order_details
Structure
Format Currency
Formatting
Data category Uncategorized
Properties
New measure
Quick measure
Calculations
1 Avg Pizza per order = [Total Pizzas Sold] / [Total Orders]
21K
Total Orders
50K
Total Pizzas Sold
$817.86K
Revenue
$38.31
Avg Order Value
$2.32
Avg Pizza per order
16.44
Avg Pizza price
9.75
Minimum Pizza Price
35.95
Maximum Pizza Price
Visualizations
Build visual
Gauge
Values
Add data fields here
Drill through
Cross-report
Off
Keep all filters
On
Add drill-through fields here
Filters
Data
Search
order_details
Avg Order Value
Avg Pizza per
order_details_id
order_id
pizza_id
quantity
Total Orders
Total Pizzas Sold
Total Revenue
ordered_amount_details
order_id
Revenue
total_amount
ordered_pizza_details
orders
pizza_details
pizza_types
pizzas


Name Total Revenue
Home table order_details
Structure
Format Currency
Auto
Formatting
Data category Uncategorized
Properties
New measure
Quick measure
Calculations
1 Total Revenue = SUMX(order_details, order_details[quantity] * RELATED(pizzas[price]))
21K
Total Orders
50K
Total Pizzas Sold
$817.86K
Revenue
Visualizations
Build visual
Filters
Data
Search
order_details
Avg Order Value
Avg Pizza per o...
order_details_id
order_id
pizza_id
quantity
Total Orders
Total Pizzas Sold
Total Revenue


Name
Avg Pizza price
Home table
pizzas
Structure
Format
Currency
Formatting
Data category
Uncategorized
Properties
New measure
Quick measure
Calculations
Avg Pizza price = AVERAGE(pizzas[price])
21K
Total Orders
50K
Total Pizzas Sold
$817.86K
Revenue
$38.31
Avg Order Value
$2.32
Avg Pizza per order
$16.44
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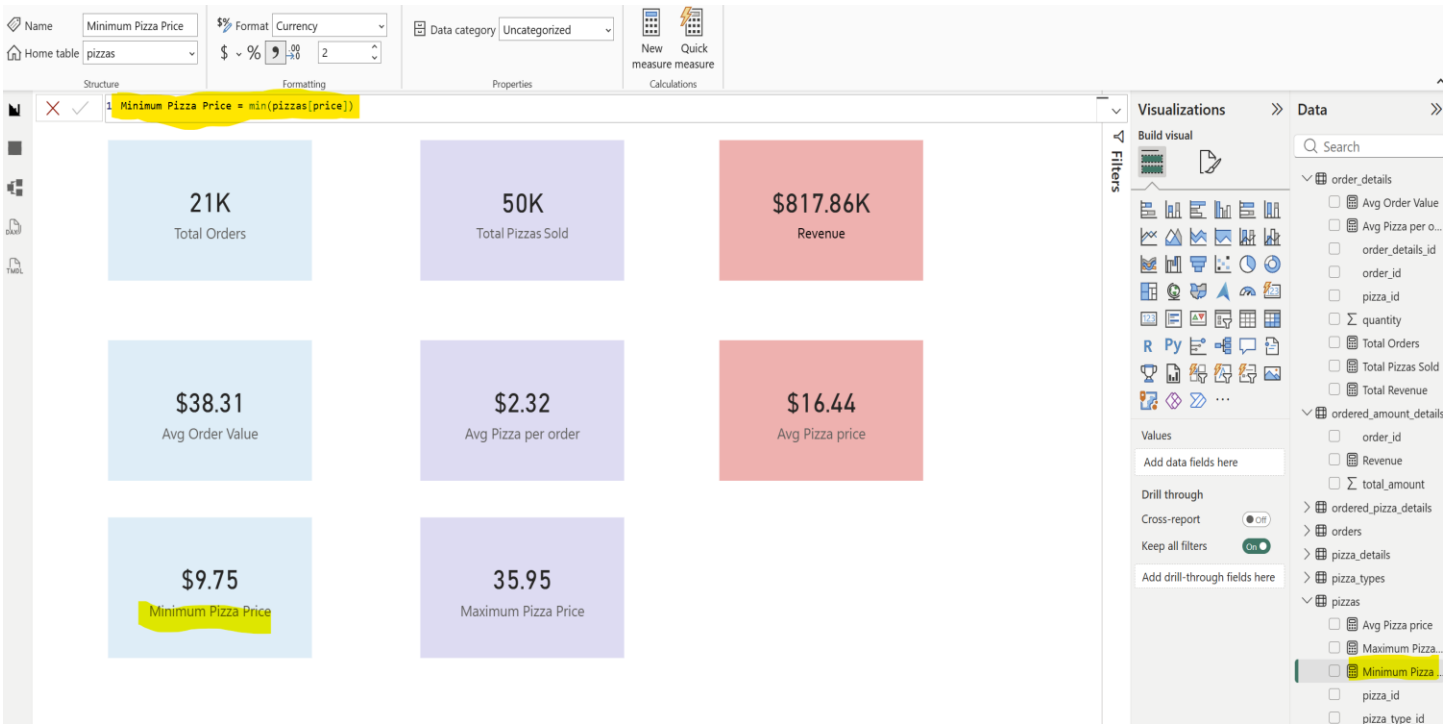
Add drill-through fields here
Data
Search
order_details
Avg Order Value
Avg Pizza per o...
order_details_id
order_id
pizza_id
quantity
Total Orders
Total Pizzas Sold
Total Revenue
ordered_amount_details
order_id
Revenue
total_amount
ordered_pizza_details
orders
pizza_details
pizza_types
pizzas
Avg Pizza pric
Maximum Pizza...
Minimum Pizza ...
pizza_id


Name
Minimum Pizza Price
Home table
pizzas
Structure
Format
Currency
Formatting
Data category
Uncategorized
Properties
New measure
Quick measure
Calculations
Minimum Pizza Price = min(pizzas[price])
21K
Total Orders
50K
Total Pizzas Sold
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Revenue
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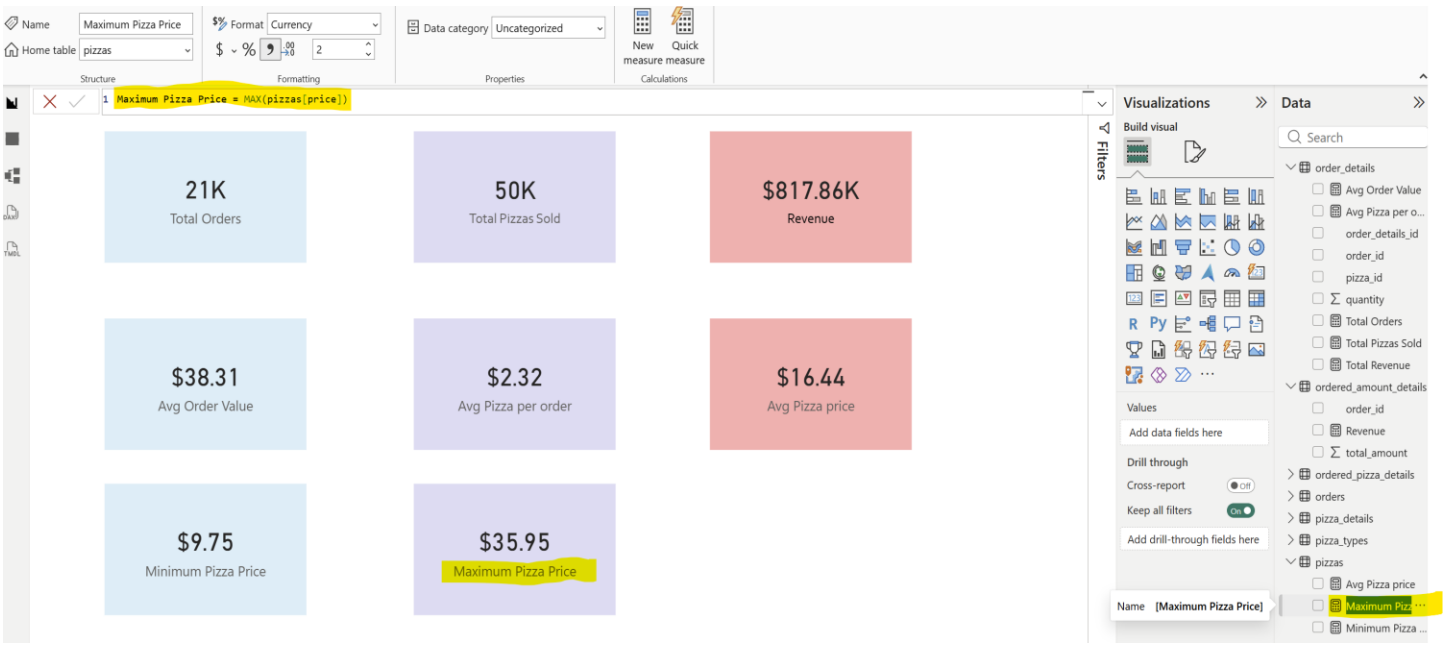
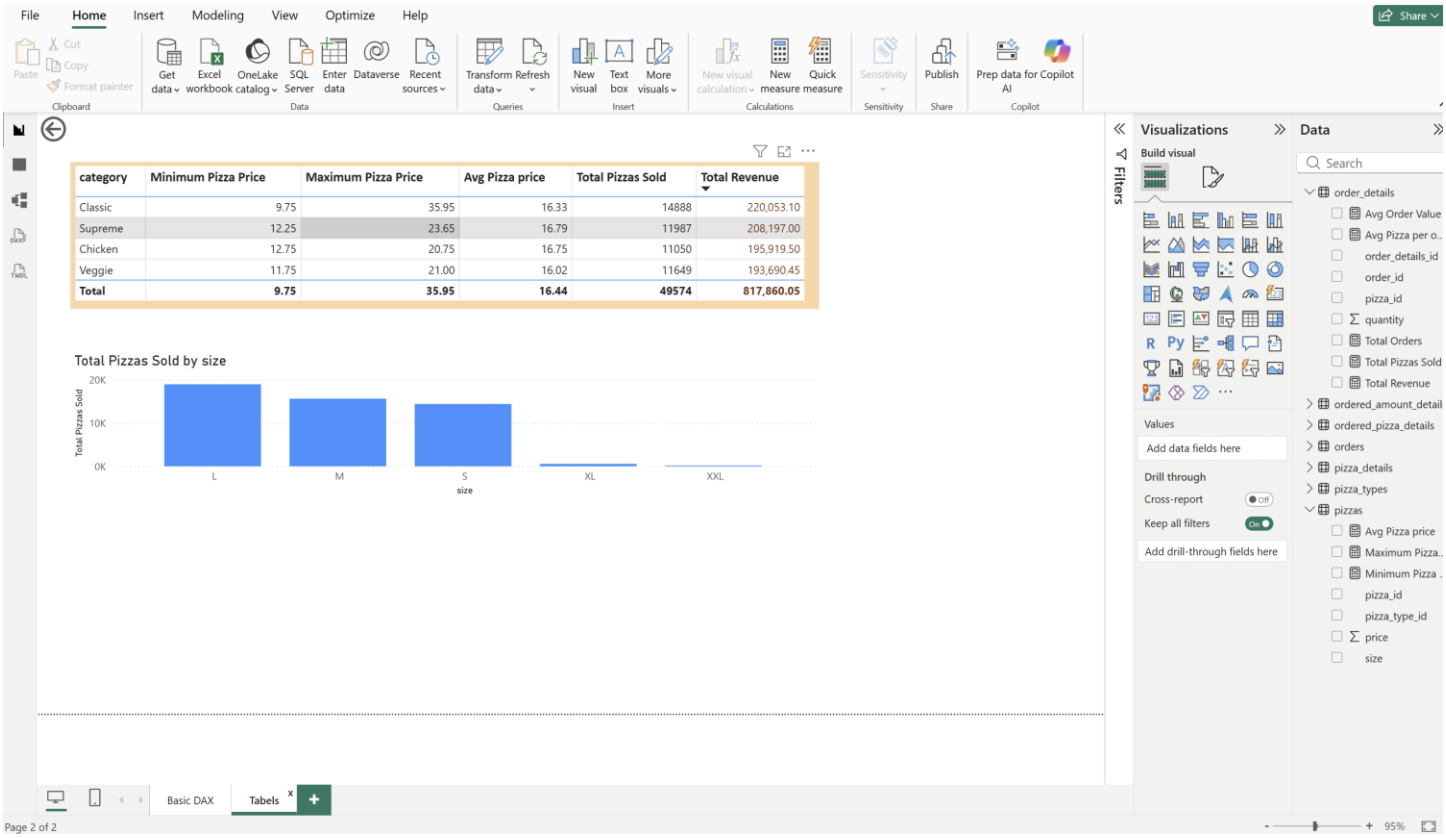
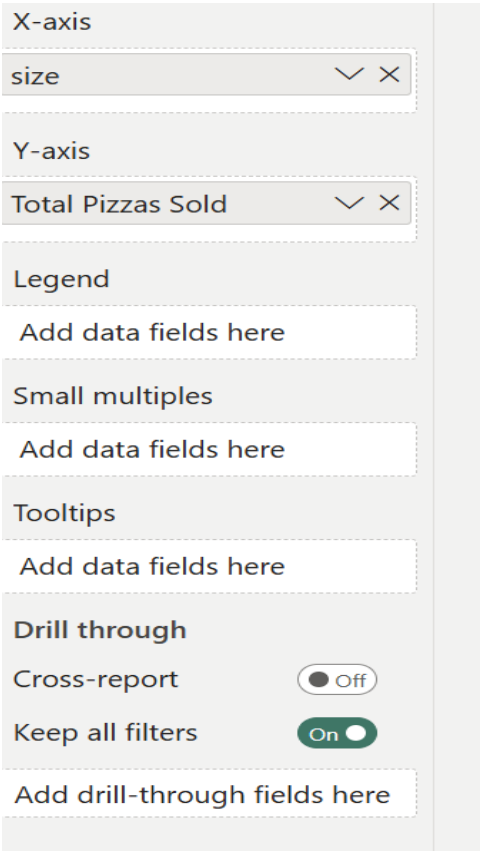
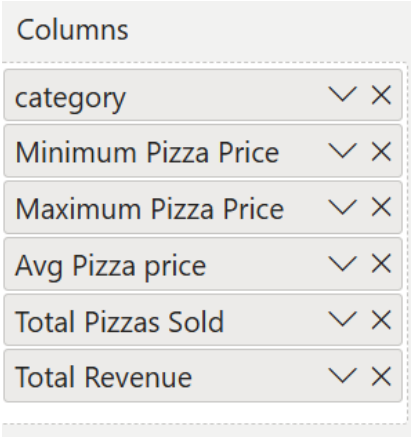
Total Revenue
ordered_amount_details
order_id
Revenue
total_amount
ordered_pizza_details
orders
pizza_details
pizza_types
pizzas
Avg Pizza price
Maximum Pizza...
Minimum Pizza ...
pizza_id
pizza_type_id





category	Minimum Pizza Price	Maximum Pizza Price	Avg Pizza price	Total Pizzas Sold	Total Revenue
Classic	9.75	35.95	16.33	14888	220,053.10
Supreme	12.25	23.65	16.79	11987	208,197.00
Chicken	12.75	20.75	16.75	11050	195,919.50
Veggie	11.75	21.00	16.02	11649	193,690.45
Total	**9.75**	**35.95**	**16.44**	**49574**	**817,860.05**





Create measures using time intelligence functions:

Total sales measure created in Order_details

```
Total Sales = SUMX(order_details, order_details[quantity] * RELATED(pizzas[price]))
```

Average Order Value measure is created

```
Average Order Value = DIVIDE([Total Sales], [Total Orders])
```

Sales Year-to-Date

```
Sales This Year =
CALCULATE([Total Sales], DATESYTD('orders'[date]))
```

Sales Month-to-Date

```
Sales This Month = CALCULATE([Total Sales], DATESMTD('orders'[date]))
```

Sales Quarter-to-date

```
Sales QTD = TOTALQTD([Total Sales], 'orders'[date])
```

Orders QTD

```
Orders QTD = TOTALQTD([Total Orders], 'orders'[date])
```

Orders Year-to-Date

```
Orders YTD = TOTALYTD([Total Orders], 'orders'[date])
```

Create measures to perform ranking and top N analysis:

Ranking Measure:

Rank pizza by total sales

```
Pizza Rank =
RANKX(
    ALL(pizzas[pizza_id]),
    CALCULATE([Total Sales]),
    ,
    DESC,
    DENSE
)
```

Rank Pizza Types by Total Sales

```
Pizza Type Rank = RANKX(
    ALL(pizza_types[name]),
    CALCULATE([Total Sales]),
    ,
    DESC,
    DENSE
)
```

Rank Categories by total sales

```
Category Rank =
RANKX(
    ALL(pizza_types[category]),
    CALCULATE([Total Sales]),
    ,
    DESC,
    DENSE
)
```

Top 5 Pizzas by Sales

```
Top 5 Pizzas Sales =
CALCULATE(
    [Total Sales],
    TOPN(5, ALL(pizzas[pizza_id]), [Total Sales], DESC)
)
```

Top 5 Pizza Types by Sales

```
Top 5 Pizza Types Sales =
CALCULATE(
    [Total Sales],
    TOPN(5, ALL(pizza_types[name]), [Total Sales], DESC)
)
```

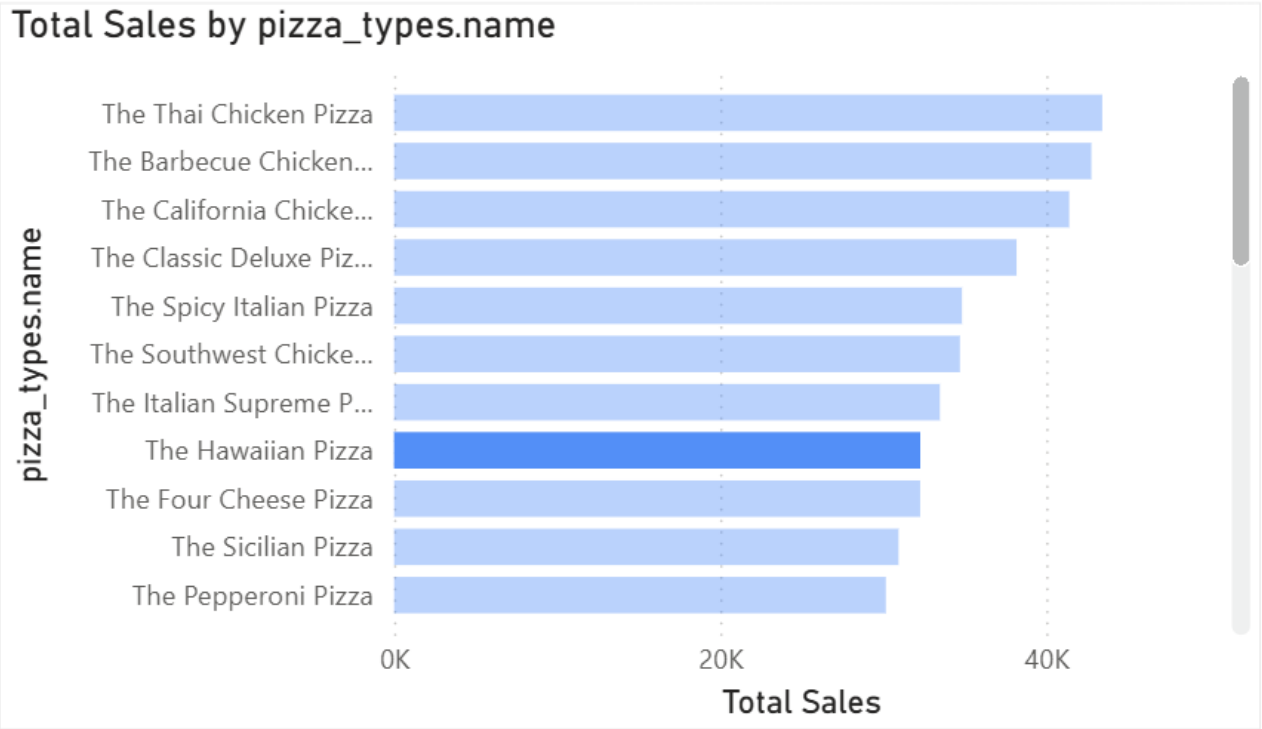
Top 5 Categories by Sales

```
Top 5 Categories Sales =
CALCULATE(
    [Total Sales],
    TOPN(5, ALL(pizza_types[category]), [Total Sales], DESC)
)
```

86.11K
Top 5 Pizzas Sales

220.05K
Top 5 Categories Sales

154.16K
Top 5 Pizza Types Sales



pizza_types.category	Total Sales	Category Rank	Top 5 Categories Sales
Chicken	195,919.50	1	195,919.50
Classic	220,053.10	1	220,053.10
Supreme	208,197.00	1	208,197.00
Veggie	193,690.45	1	193,690.45
Total	**817,860.05**	**1**	**817,860.05**

PIZZA SALES KPIS & INSIGHTS DASHBOARD:

- Overall Performance: 21K total orders generated $817.86K in revenue, with an average of 2.32 pizzas per order.
- Category Trends: Classic pizzas dominate with ~29% of orders and ~27% of sales, showing strong customer preference.
- Balanced Distribution: Supreme, Chicken, and Veggie categories each contribute around 23–25%, keeping sales spread evenly.
- Pricing Insight: Average pizza price is $16.44, ranging from $9.75 to $35.95, reflecting a wide product mix.
- Top Sellers: The Sicilian and Spicy Italian pizzas stand out among individual types, driving notable category sales.